

CAPRM-Flood: Flood Exposure Evidence

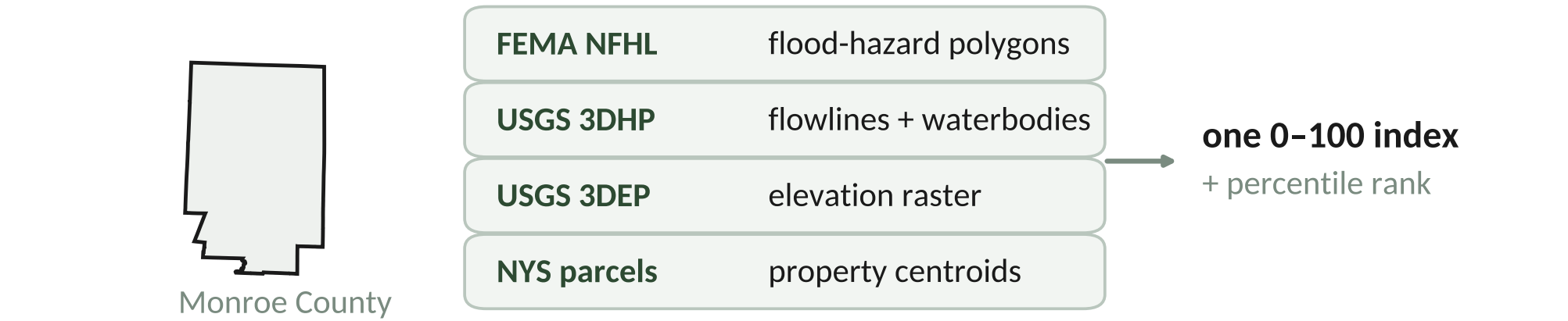
Climate-Adjusted Property Risk Modeler · 267,362 properties, one exact answer each. What does exactness cost, and can it be avoided?

ABSTRACT

CAPRM-Flood extracts traceable, property-level flood evidence from public geospatial data for every parcel in Monroe County, New York, and combines it into a transparent relative exposure score. The dominant computation is an exact nearest-water distance, repeated 267,362 times over 1,063,159 pieces of geometry. This work asks what that exactness costs, whether an index can reduce it without approximating the answer, and whether it can be avoided altogether.

THE SYSTEM

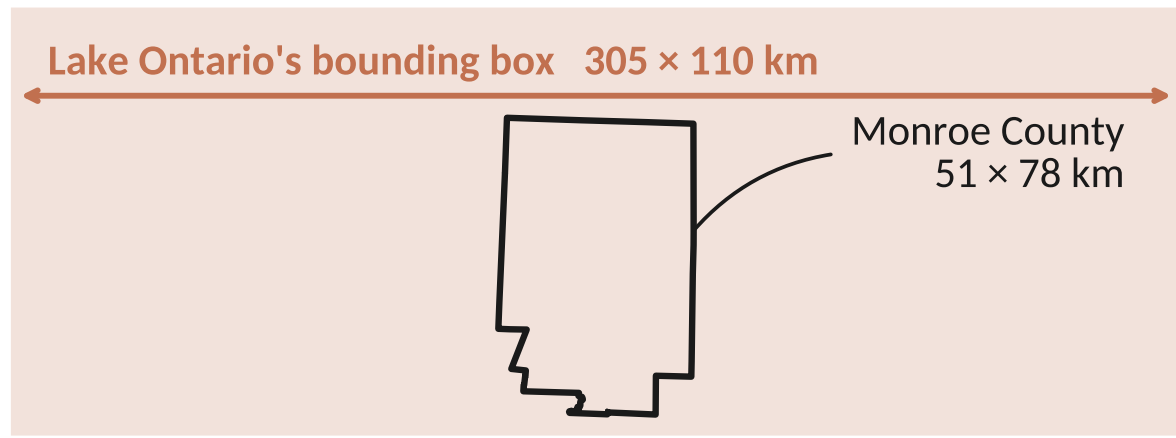
Four public datasets, one number per property



Four evidence values per property: mapped flood-hazard zone, distance to the nearest water feature, ground elevation, and elevation relative to surrounding terrain. Four declared weights combine them into a 0 to 100 exposure score, interpreted as a rank within the county. Nearest-water distance is the only value that must be measured rather than sampled. Water is 5,600 flowlines and 2,972 waterbodies, each a chain of straight two-point pieces, 1,063,159 in total. One exact answer measures a property against all of them: 284 billion comparisons countywide, 1,139.7 s by brute force.

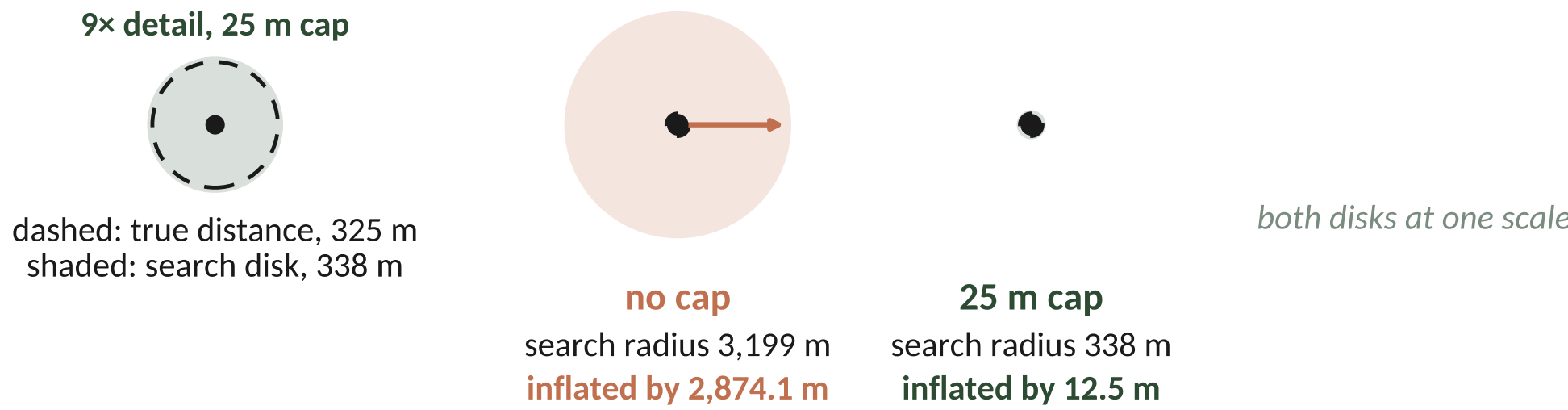
CAN THE DOMINANT QUERY BE MADE CHEAPER WITHOUT LOSING EXACTNESS?

A box covering the county can never be pruned



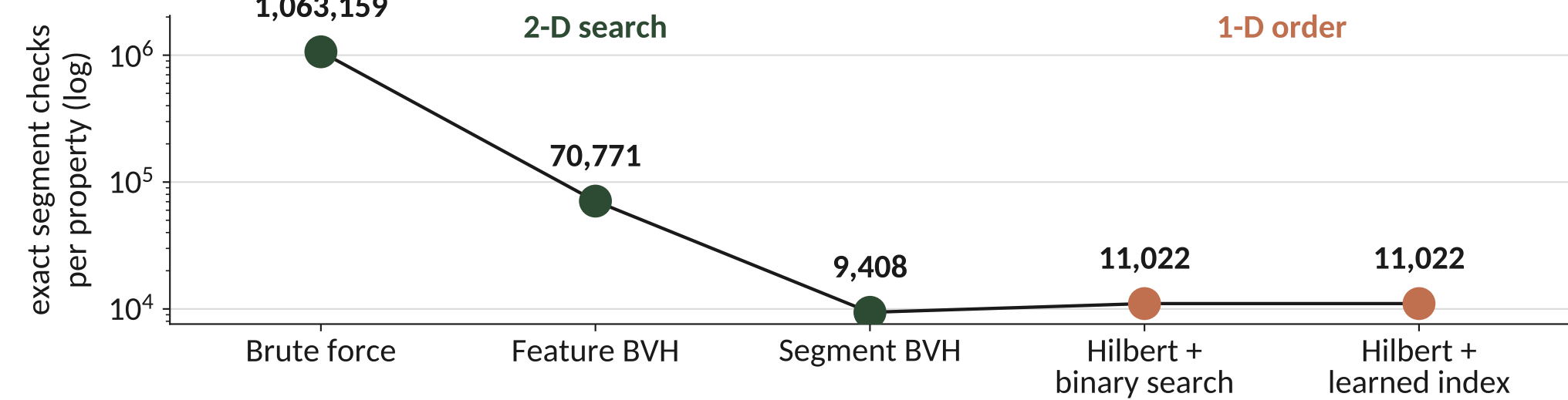
A spatial index skips comparisons it can prove cannot win: geometry is wrapped in axis-aligned boxes, and any box farther away than the best distance so far is discarded with its contents unexamined. Over the 8,572 whole features, about 5 survive per property and 93.34% of exact checks disappear. 70,771 remain. Lake Ontario's box spans 305 by 110 km and contains every property in the county, so its distance to every query is zero and it is never pruned. A feature is the smallest unit this hierarchy resolves. Indexing the 1,063,159 pieces instead cuts checks to 9,408.

Sorting extended objects widens the search



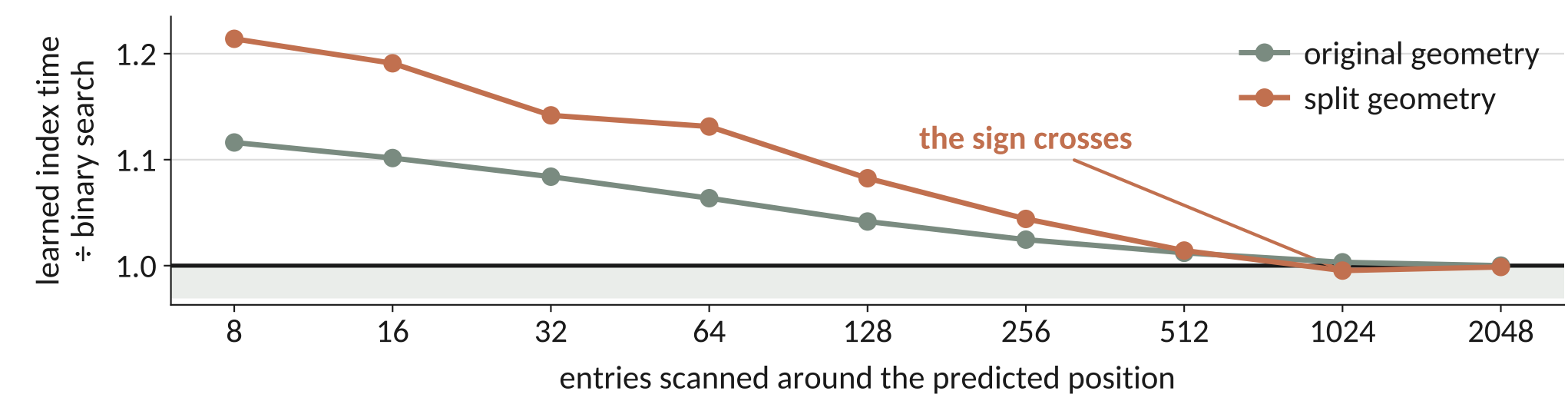
A learned index predicts where a key sits in a sorted array. Coordinates cannot be sorted, so a Hilbert curve assigns every location a single number. Sorting stores each piece at its midpoint and discards its extent. A midpoint 2,000 m away can own an endpoint 10 m away. Exactness therefore requires admitting every midpoint within half the longest piece: 2,874 m. Capping piece length at 25 m reduces that to 12.5 m for 11.89% more entries. No distance changes, because a split piece lies on its parent.

Same answer, less work



The kernel, the exact point-to-piece distance calculation, is identical in all five implementations. Only the choice of which pieces reach it changes. Exact checks per property: 1,063,159 by brute force, 70,771 over feature boxes, 9,408 over piece boxes, 11,022 over the Hilbert order. Flattening to one dimension costs 17.2 percent back, because an order prunes less well than a tree. All five agree with brute force on all 267,362 properties, maximum error 4.658×10^{-10} m.

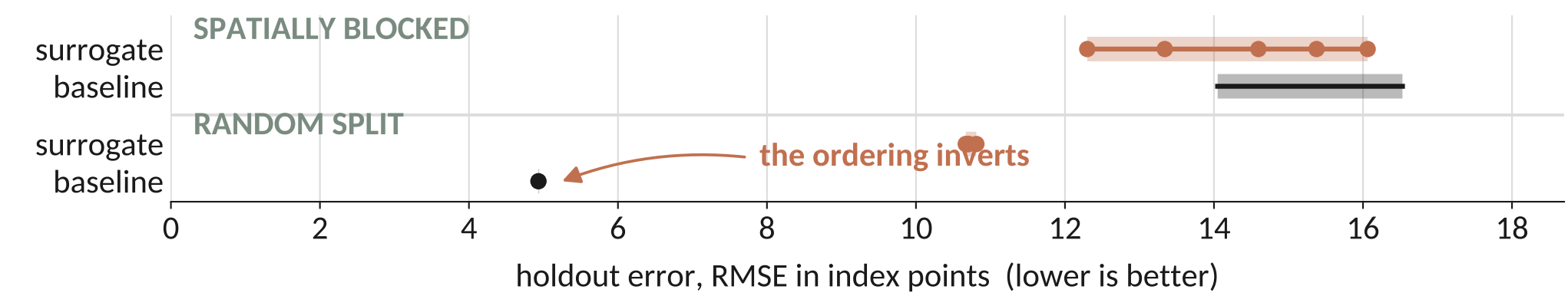
The search window decides which method wins



Binary search locates a key in 20.2 probes. The learned index predicts the position and corrects itself over a window of W neighbouring entries, in 6.3. W is not reported in the learned-index literature. Across its range, 8 to 2048, the time ratio moves from 1.21 to 1.00 and crosses. Both implementations return identical distances. The learned index reduces search; verification dominates the total.

CAN IT BE REMOVED INSTEAD?

Random splits measure interpolation



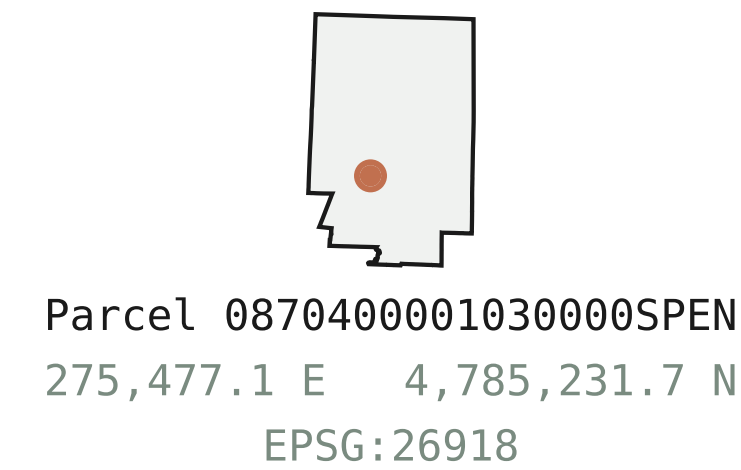
A neural network predicts the exposure score from coordinates alone, computing no geometry. The baseline, declared in writing before training, copies the score of the nearest training property. Test properties are withheld in 10 km blocks separated from training data by 2,125 m, the lag at which half the spatial correlation is gone. Blocked, the network does not clear the baseline. Split at random, it does.

The exact pipeline stands. Exactness over extended objects has a price, 11.89% more entries, and paying it costs 113x less work than brute force.

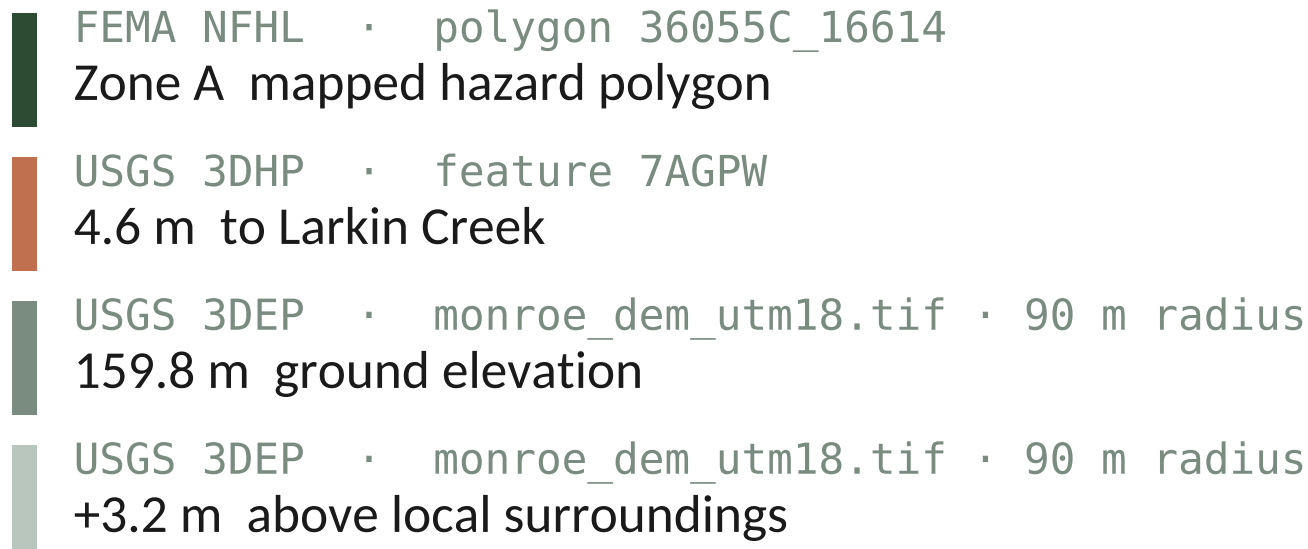
THE OUTPUT: ONE PROPERTY, END TO END

every number traceable to a named source; every artifact checksummed

WHERE IT IS



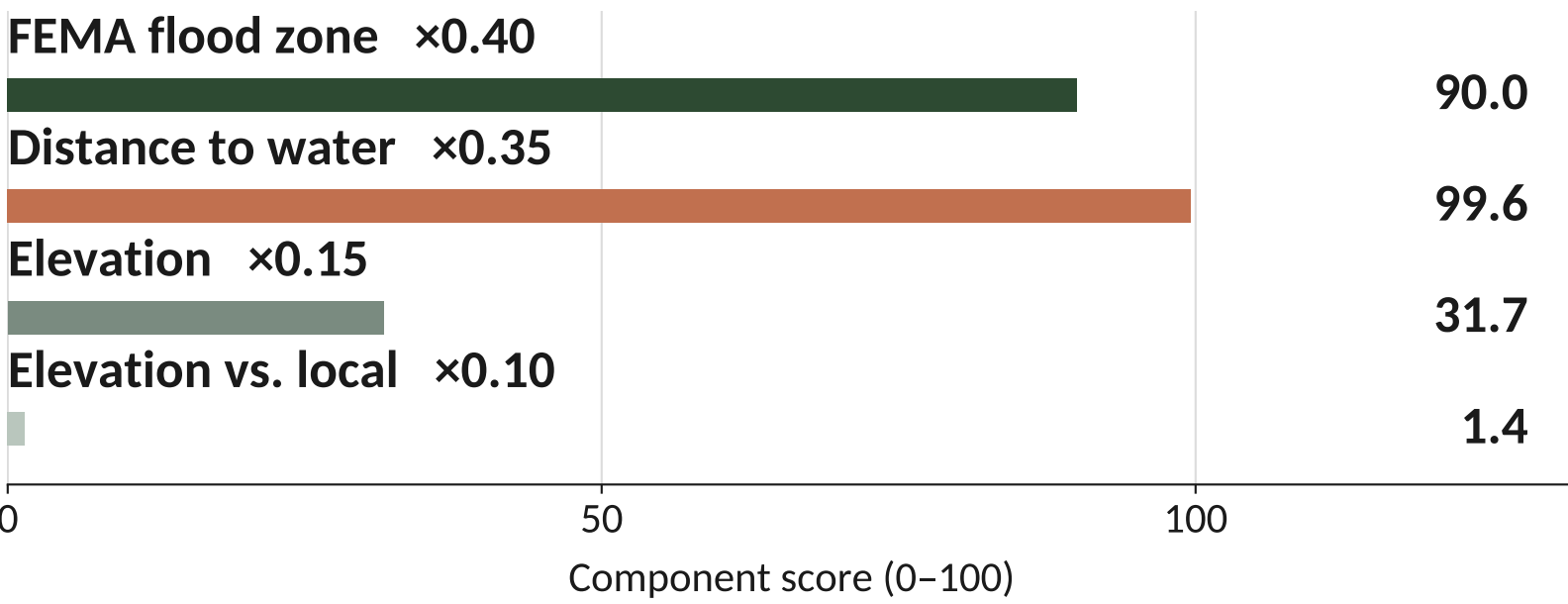
WHAT WAS MEASURED



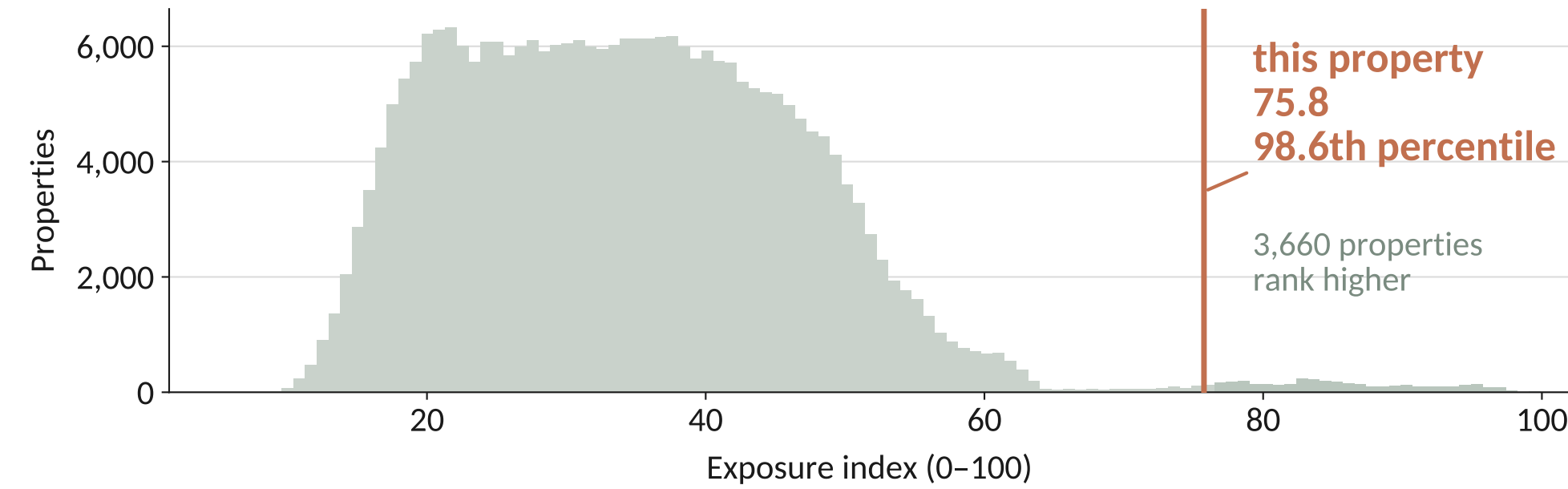
$$0.40 \times 90.0 + 0.35 \times 99.6 + 0.15 \times 31.7 + 0.10 \times 1.4 = 75.8$$

The whole index is this line. FEMA is an absolute lookup; the other three rank this property against all 267,362 in the county. Recomputing the composite from the four components and the four declared weights reproduces the stored value to 2.3e-12.

HOW IT SCORED



WHERE IT LANDS



CORRECTNESS

Computed twice, Python and independent C++. All 267,362 agree to 4.658×10^{-10} m. Ranking holds at Spearman ≥ 0.875 .

REPRODUCIBILITY

653 Python tests pass. C++ suites run 80,021 and 607 checks. Artifact audit 49 pass, 1 warn, 0 fail. Manifests and checksums throughout.

SOURCES

FEMA NFHL · USGS 3DHP · USGS 3DEP · NYS parcel centroids · TIGERweb. Checksummed; NFHL and DEM retrieval dates unrecorded.

CRS AND SCOPE

EPSG:26918 throughout, never degrees. The score is a relative rank within Monroe County, not a flood probability, an expected loss, or an insurance rating.